

Datadog Observability Analysis Report

Dataset: staging_input | **Generated:** 2026-07-10T11:26:34Z **Analysis Period:** 2026-07-15T14:00:00.000Z -> 2026-07-15T15:02:01.000Z **Overall Health:** CRITICAL

Executive Summary

- Total incidents identified: 2
- Critical issues: 14 | Error issues: 7 | Warnings: 5
- Top risks:
 - video-encoder identified as origin; failure propagated to cdn-gateway, recommendation-engine within 5 minutes
 - host enc-host-01 flagged HIGH_DISK (verdict CRITICAL)

1. Errors & Data Quality

Service	Issue	Severity	Detail
video-encoder	RESOURCE_EXHAUSTION	CRITICAL	Disk usage on /var/encode/tmp reached 98%, encoder pool degraded
video-encoder	RESOURCE_EXHAUSTION	CRITICAL	Disk usage on /var/encode/tmp reached 99%, encoder pool offline
video-encoder	RESOURCE_EXHAUSTION	ERROR	Encode job failed for asset_id=9932: No space left on device
video-encoder	RESOURCE_EXHAUSTION	ERROR	Encode job failed for asset_id=9933: No space left on device
video-encoder	RESOURCE_EXHAUSTION	ERROR	Encode job failed for asset_id=9934: No space left on device

...2 more error(s) not shown; see log_analysis.json.all_errors for the full list.

No data-quality rejection evidence found in this analysis period.

2. Performance & Infrastructure

Latency by service:

Service	Avg (ms)	P95 (ms)	P99 (ms)	Verdict
cdn-gateway	2683.8	5285.0	5297.0	CRITICAL
video-encoder	2623.8	5193.5	5206.7	CRITICAL
recommendation-engine	4800.0	4800	4800	CRITICAL

Service	Avg (ms)	P95 (ms)	P99 (ms)	Verdict
billing-service	200.0	200	200	OK

Slowest traces:

Service	Operation	Duration (ms)	Timestamp
cdn-gateway	fetch_asset	5300	2026-07-15T14:27:00.000Z
video-encoder	encode_request	5210	2026-07-15T14:27:01.000Z
cdn-gateway	fetch_asset	5200	2026-07-15T14:26:00.000Z
video-encoder	encode_request	5100	2026-07-15T14:26:01.000Z
recommendation-engine	get_recommendations	4800	2026-07-15T14:27:02.000Z

Host resource health:

Host	CPU %	Memory %	Disk %	Verdict	Issues
enc-host-01	91.0	74.0	99.0	CRITICAL	HIGH_CPU, HIGH_DISK, HOST_DOWN
cdn-host-01	60.0	48.0	36.0	CRITICAL	NETWORK_SATURATION, HOST_DOWN
bill-host-01	22.0	35.0	28.0	WARN	HOST_DOWN
rec-host-01	28.0	38.0	30.0	WARN	HOST_DOWN

All performance/infra issues:

Service	Issue	Severity
cdn-gateway	HIGH_ERROR_RATE	CRITICAL
video-encoder	HIGH_ERROR_RATE	CRITICAL
recommendation-engine	HIGH_ERROR_RATE	CRITICAL
enc-host-01	HIGH_CPU	CRITICAL
enc-host-01	HIGH_DISK	CRITICAL
enc-host-01	HOST_DOWN	CRITICAL
cdn-host-01	NETWORK_SATURATION	CRITICAL
cdn-host-01	HOST_DOWN	CRITICAL
cdn-gateway	HIGH_LATENCY	CRITICAL
video-encoder	HIGH_LATENCY	CRITICAL
recommendation-engine	HIGH_LATENCY	CRITICAL

...2 more issue(s) not shown; see metrics_report.json.all_issues for the full list.

3. Pipeline Health

No Kafka consumer lag metrics or log evidence were found in this analysis period.

All pipeline issues:

Service	Issue	Severity
video-encoder	PIPELINE_ALERT_UNCLASSIFIED	● CRITICAL
video-encoder	PIPELINE_ALERT_UNCLASSIFIED	● ERROR
cdn-gateway	PIPELINE_ALERT_UNCLASSIFIED	● ERROR

4. Security

No security findings (PII exposure, credential leaks, unauthorized access) detected in this analysis period.

5. Anomalies & Trends

Service	Anomaly	Confidence	Detail
enc-host-01	CPU_SPIKE	MEDIUM	enc-host-01 cpu_pct reading 88.0 exceeded 2.0x its own prior baseline 42.5

6. Dependency & Breakpoint Analysis

Service graph: 3 node(s), 2 edge(s).

Breakpoint Service	Downstream Impact	Confidence
video-encoder	cdn-gateway, recommendation-engine	0.65

Root Cause Analysis

incident_004 — **RESOURCE_SATURATION** (confidence: HIGH, severity: CRITICAL, blast radius: 3)

- Primary service: video-encoder
- Root cause: video-encoder identified as origin; failure propagated to cdn-gateway, recommendation-engine within 5 minutes
- Downstream symptoms: Encode job failed for asset_id=9932: No space left on device; Encode job failed for asset_id=9933: No space left on device; Disk usage on /var/encode/tmp reached 98%, encoder pool degraded; Disk usage exceeded 95% on enc-host-01; Encode job failed for asset_id=9934: No space left on device; encode queue depth exceeded 500 on video-encode-queue; Upstream timeout: video-encoder not responding within 5000ms; cdn-gateway error rate 50.0% across 4 traces; cdn-gateway p99 latency 5297.0ms; Upstream timeout rate exceeded threshold; video-encoder error rate 50.0% across 4 traces; video-encoder p99 latency 5206.7ms; recommendation-engine error rate 100.0% across 1 traces; recommendation-engine p99 latency 4800ms; Disk usage on /var/encode/tmp reached 99%, encoder pool offline
- Evidence sources: apm_report.json, dependency_report.json, log_analysis.json, metrics_report.json

incident_005 — **RESOURCE_SATURATION** (confidence: MEDIUM, severity: CRITICAL, blast radius: 1)

- Primary service: enc-host-01
- Root cause: host enc-host-01 flagged HIGH_DISK (verdict CRITICAL)
- Downstream symptoms: host enc-host-01 flagged HIGH_CPU (verdict CRITICAL); host enc-host-01 flagged HOST_DOWN (verdict CRITICAL); enc-host-01 cpu_pct reading 88.0 exceeded 2.0x its own prior baseline 42.5
- Evidence sources: anomaly_report.json, metrics_report.json

⚠️ 2 critical finding(s) could not be clustered into an incident -- see `root_cause.json.unresolved_findings`.

Recommendations

Rank	Priority	Title	Affected Services
1	P1_IMMEDIATE	Address RESOURCE_SATURATION on video-encoder	cdn-gateway, recommendation-engine, video-encoder
2	P1_IMMEDIATE	Address RESOURCE_SATURATION on enc-host-01	enc-host-01

Patch Suggestions (Human Review Required)

⚠️ All patches require human review before applying.

patch_001 (LOW risk, **CONNECTION_POOL_RESIZE**) — targets `config/connection-pool.yaml`

- enc-host-01 showed sustained resource exhaustion; double the connection/thread pool size to relieve pressure (confirm current pool_size before applying).

```
- pool_size: 20 # enc-host-01
+ pool_size: 40 # enc-host-01
```

1 recommendation(s) flagged for manual review instead of an automated patch.

Appendix — Ingestion Summary

Source Type	Record Count
log	14
metric	26
trace	10
alert	4

Source Type	Record Count
infrastructure	8

Total records ingested: 62